

## 📁 01PYTHON\_VAULT/

|  
|— 📁 00\_CHEATSHEETS&SNIPPETS/ <-- Quick syntax, virtual env setups, reusable functions

|— 📁 01\_DATA\_SCIENCE&ANALYTICS/ <-- Data handling & visualization

| |— 📁 Pandas/ <-- DataFrames, CSV/Excel parsing, data cleaning

| |— 📁 NumPy/ <-- Array operations, mathematical functions

| |— 📁 Matplotlib\_Seaborn/ <-- Data visualization & plotting

| |— 📁 Polars/ <-- Fast DataFrame alternative

|— 📁 02\_AI&MACHINE\_LEARNING/ <-- AI engineering & model training

| |— 📁 PyTorch/ <-- Deep learning models & neural networks

| |— 📁 OpenAI\_API/ <-- API integrations, embeddings, custom functions

| |— 📁 LangChain\_LlamaIndex/ <-- RAG pipelines & LLM orchestration

| |— 📁 HuggingFace/ <-- Transformers, pre-trained models, datasets

| |— 📁 Scikit\_Learn/ <-- Traditional ML algorithms (regression, trees)

|— 📁 03\_WEB\_FRAMEWORKS/ <-- Backend API & Web development

| |— 📁 FastAPI/ <-- Modern async APIs & OpenAPI specs

| |— 📁 Django/ <-- Full-stack web applications

| |— 📁 Flask/ <-- Micro-frameworks & quick endpoints

|— 📁 04\_AUTOMATION&SCRAPING/ <-- Web scraping & task automation

|— 📁 Playwright\_Selenium/ <-- Browser automation

|— 📁 BeautifulSoup/ <-- HTML parsing & web scraping

|— 📁 Requests\_Aiohttp/ <-- HTTP networking & async fetching

## 📁 DEVWORKSPACE/

|— 📁 00\_ACTIVE\_PROJECTS/ # Everything you are actively building right now

| |— 📁 WEB/ # Active web applications

| |— 📁 APP/ # Active mobile apps

| |— 📁 AI\_ML/ # Active AI experiments & pipelines

|— 📁 01\_PYTHON\_VAULT/ # Centralized Python knowledge base

| |— 📁 00\_Cheatsheets&Snippets/

| |— 📁 01\_Data\_Science&Analytics/

| | |— 📁 Pandas/

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| | └─  NumPy/
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| | └─  Polars/
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| | └─  Scikit_Learn/
| └─  03_Web_Frameworks/
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| └─  BeautifulSoup/
| └─  Requests_Aiohttp/
|
└─  02_LEARNING&RESEARCH/ # Study materials & research
| └─  Online_Courses/ # Interactive web courses
| └─  Video_Courses/ # Video tutorials & downloads
| └─  Research&_Notes/ # Articles, PDFs, AI papers
|
└─  03_ARCHIVE/ # Finished projects go here
└─  COMPLETED_PROJECTS/ # Moved out of 00_ACTIVE_PROJECTS
└─  PAST_COURSES/ # Completed study materials

```

### Workspace Directory Map

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└─ 01_PYTHON_VAULT/
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|   └─ 01_Data_Science_&_Analytics/
|       └─ Pandas/
|       └─ NumPy/
|       └─ Matplotlib_Seaborn/
|       └─ Polars/
└─ 02_AI_&_Machine_Learning/

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| | | └─ LangChain_LlamaIndex/
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